PART 1.

Assessment of Stevens Flood Advisory System (SFAS) service and predictions for Spring Tidal Event of April 1st 2017

A total of 34 SFAS stations experienced mostly Minor Flooding during a Spring Tidal Event on April 1st, 2017 (Table 1). Four days before the event, between 9am and 12pm EST March 27th 2017, automated Flood Watch notices were sent to SFAS subscribers, warning them of a coming flood event. Most notices were followed next day with a 3-day out flood watch notice.

The automatically -mailed Flood Watch message from SFAS read:

There is a possibility of significant flooding (water levels rising above moderate flood stage) for the station(s) listed below 4 days out from now. We suggest that you use the link for each station below to check the predictions, updated every 6 hours.

Table 1: 34 flooded stations during the Spring Tidal Event on April 1 2017. SFAS 4-day Flood Watches for possible moderate flooding are highlighted with italics. Watches for which a 3-day Flood Advisory was also issued are highlighted in Bold to show consistency of early forecasts. Short-term Flood Advisories issued in advance of peak heights the day of the storm, April 1, 2017, are underlined. Automated Early Flood Watches were issued for a total of 8 coastal stations, with 28-station in Automated Flood Advisory e-mails the day of the storm. URL color shows the verifying NWS local flood conditions: Above Minor (Yellow), Above Moderate Flood Level (Orange).

Absecon Creek Rte 30 at Absecon NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U228
Atlantic City NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N022
Barge Marina at Carlstadt NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=MEEMS_4
Bergen Point West Reach NY: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N019
Boston MA: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N065
Bowery Bay at East Elmhurst NY: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=SPLGAAP1
Cape May NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N023
Cohancey River at Greenwich NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U239
East Thorofare at Ship Bottom NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U226
FDU Bridge at Hackensack NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=MEEMS_1
Hackensack River at Hackensack NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U215
Hudson Bay at Freeport NY: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U205

Hudson River at South Dock at West Point NY: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U202

Hudson River below Poughkeepsie NY: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U201

Hudson River near Jersey City NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=SPNJTLP1

Inside Thorofare Rte 40 Atlantic City NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U229

Jamaica Bay at Inwood NY: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U208

Kearny Point at Kearny NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=MEEMS_3

Kings Point NY: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N016

Lewes DE: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N030

Little Egg Inlet near Tuckerton NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U227

Maurice River at Bivalve NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U238

New Haven CT: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N010

New London CT: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N008

Norrie Point NY: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=HRNOPTH

Passaic River at PVSC at Newark NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U216

Raritan Bay at Keansburg NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U218

Raritan River at South Amboy NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U217

Sandy Hook NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N021

Saw Mill Creek at Kearny NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=MEEMS_2

Shark River at Belmar NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U220

Shrewsbury River at Sea Bright NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U219

Tivoli Bays South NY: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=HRTVBSH

Watson Creek at Manasquan NJ: http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=U249

SFAS Flood Advisory Notices were automatically sent to subscribers at 6am and 12pm EDT, April 1st. Stations for which these short-term Flood Advisories were sent are underlined on Table 1 at the top of the document. Flood Advisories were sent for a total of 28 stations. As seen on Table 1, Flood Watches for possible above moderate flooding had also been issued for 8 stations three or four days in advance of the storm. 20 other stations received Flood Advisories the day of the storm, but no early Flood Watches (Table 1). Thus, a total of 28 out of 76 SFAS coastal stations received a Flood Watch or Advisory. For many stations, the 5%-to-95% ensemble uncertainty shown in SFAS four days prior to the event enveloped the event's local observed peak water level. However, this event was overall under-predicted early on: The 4-day-forecast cross-station average probability of exceedance of the verified April 1st event peaks was 98% (the event verified at the 2% percentile of the 96hour early prediction envelope). For some stations, like Kings Point NY and New Haven CT, the ensemble uncertainty only started enveloping the peak that was later observed 48 hours prior to April 1st, which is too late for SFAS early SFAS Flood Watches to be sent. However, as the event came close and the ensemble converged around it, automated short-term SFAS Flood Advisories were sent out in advance.

An example of the e-mailed SFAS Flood Advisories and the information they contained is given below:

We expect flooding for the station(s) listed below at some point in the eight-hour period between 2017-04-01 12:00 AM and 2017-04-01 8:00 AM local time. We have indicated the approximate time when we expect the first event to occur for each station. Note that there may be higher flooding later in the eight-hour period, and we suggest that you click on the link for each station below to see complete flooding information for that station.

Kings Point NY at approximately: 2017-04-01 1:35 AM
http://hudson.dl.stevens-tech.edu/sfas/d/index.shtml?station=N016

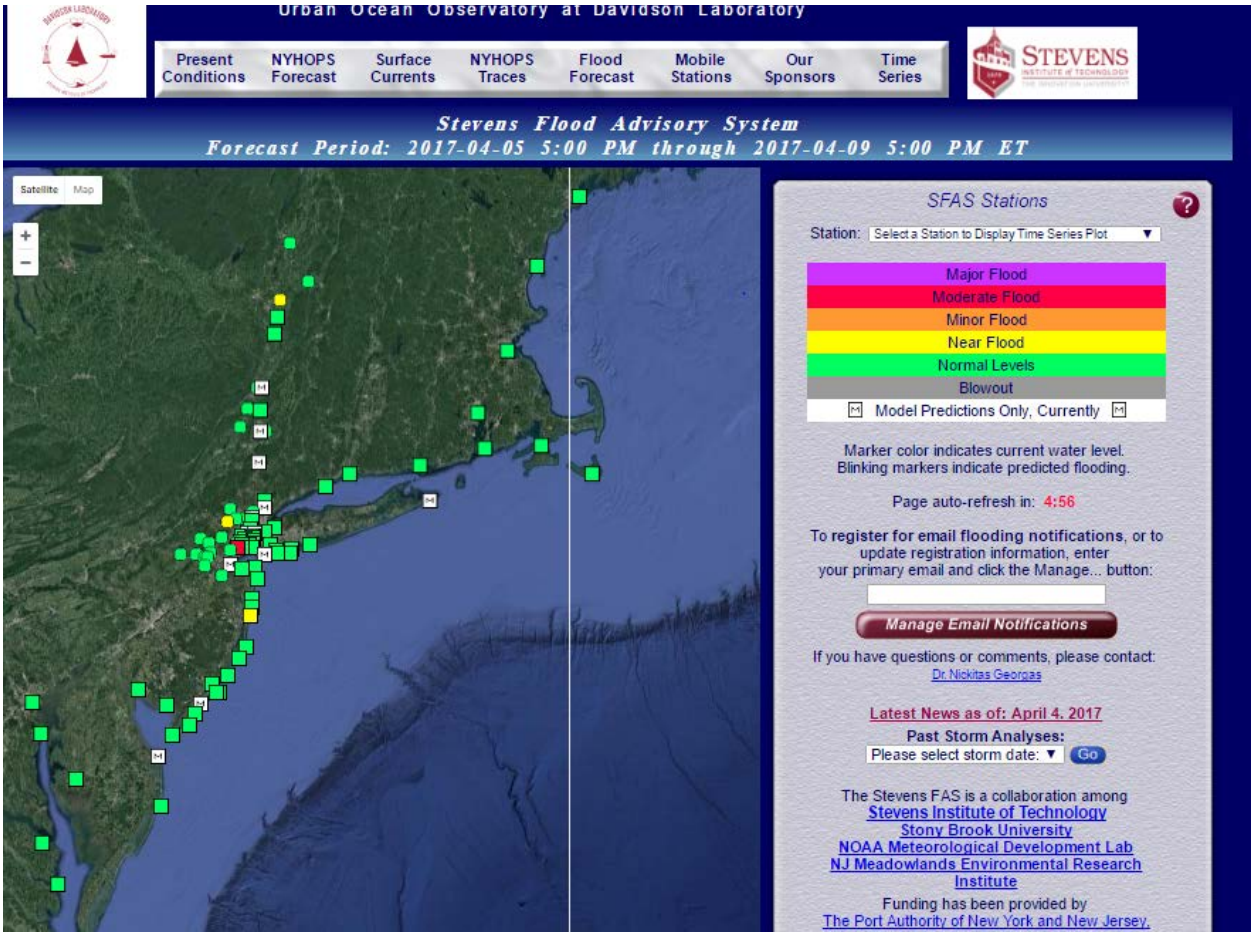
Of the 28 issued Flood Advisories, 24 verified as minor floods, while 4 verified as moderate floods; for 2 of the latter, an early Flood Watch was sent 3 days in advance, so ample warning existed. Three more stations listed on Table 1 were misses: Minor flooding was observed at Absecon Creek Rte 30 at Absecon NJ, Cape May NJ, and Lewes DE; Minor Flood Levels were exceeded by 0.3ft, 0.3ft and 0.1ft respectively. Three more near-misses were at Boston MA, Inside Thorofare Rte 40 Atlantic City NJ and New London CT where levels just exceeded Minor Flood, by less than 0.1ft. A summary is found on Table 2. These 6 misses were less than 0.3ft above Minor Flood levels and did not experienced above Moderate-level flooding, which means that the early predictions were partially verified.

Table 2. Summary of SFAS Flood Advisories for the Spring Tidal Event

Hits: 28 (stations that received early flood watches or day-of-the-storm advisories, and flooded)
False Alarms: 0 (received advisory but no flood occurred).
Misses: 6 (stations flooded, but no advisory was issued).

PART 2.

Snapshots of storm surges experienced during Spring Tidal Event April 1st 2017



Sample Stations:

- The Battery NY
- Kings Point NY
- Sandy Hook NJ
- Atlantic City NJ
- New Haven CT
- Boston MA
- Newport RI
- Lewes DE

How to read the graphs of the Stevens Flood Advisory System

In the figure for each station:

- **The graph on top left corner** includes basic information for each station, such as station's name, start date and end date, datum, forecast model displayed, unit and time zone.
- **The graph on bottom left corner** is the map of the location of the selected station.
- **The graph on top right corner shows water level information:**

~Predicted water level based on Stevens Ensemble Forecast models (line)

~Ensemble uncertainty from 5% to 95% (Gray area)

~Observed water level (Red dots, where available)

~Time marker when the snapshot was taken (vertical black dash line)

Note: For Boston MA station, the Predicted water level is based on Stevens SNAP Extended Ensemble Forecast model (dark green line); the Stevens NYHOPS Ensemble is used for the rest of the stations (pink line)

- **The graph on bottom right corner shows storm surge information:**

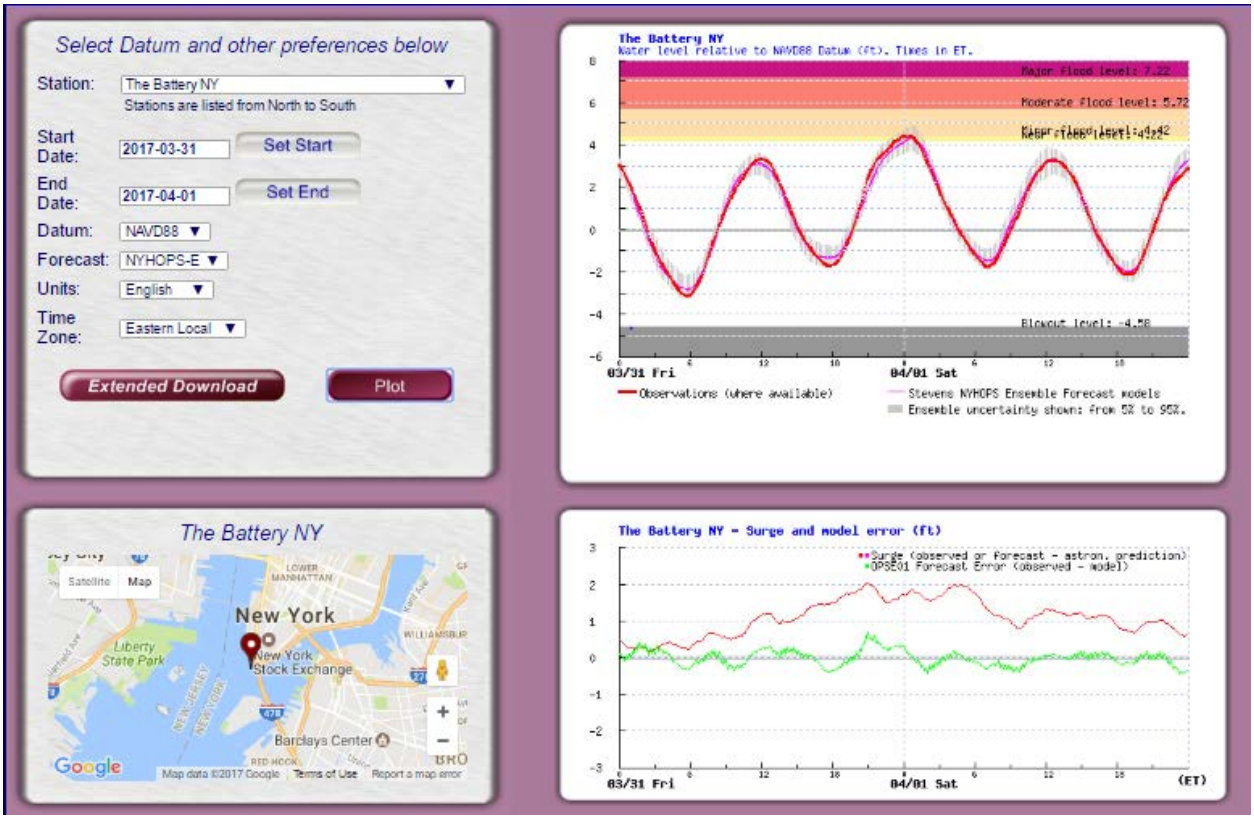
~Observed surge = observed water level – predicted astronomical tide (Red line)

~Forecast surge = predicted water level – predicted astronomical tide (Pink line)

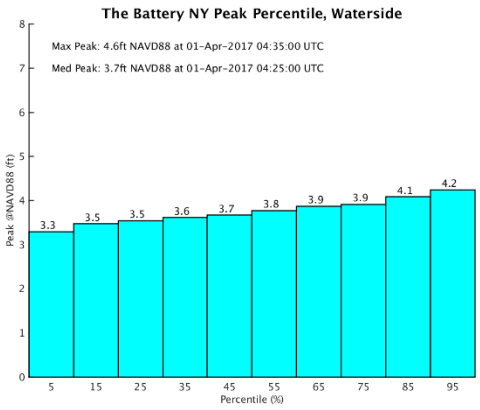
~Forecast error = observed surge – model surge (Green line)

~Time marker when the snapshot was taken (vertical black dash line)

Figures for sample stations:

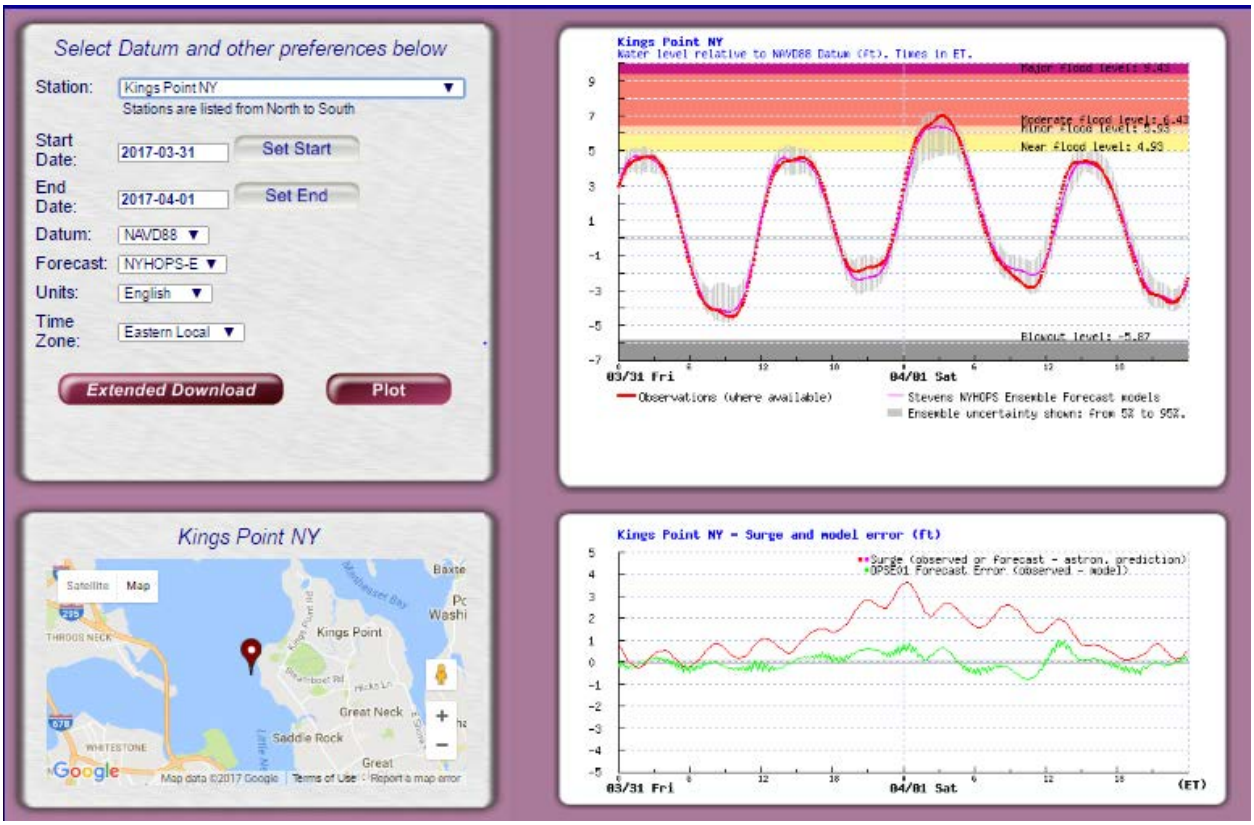


The Battery NY

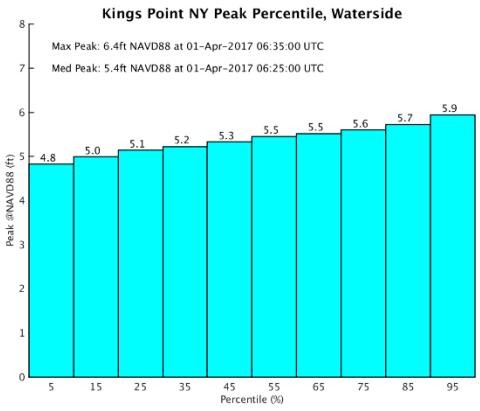


Percentiles of 96hr-prior forecast peaks.

Verified peak was 4.45ft NAVD88 on April 01 2017, 3:54 UTC.

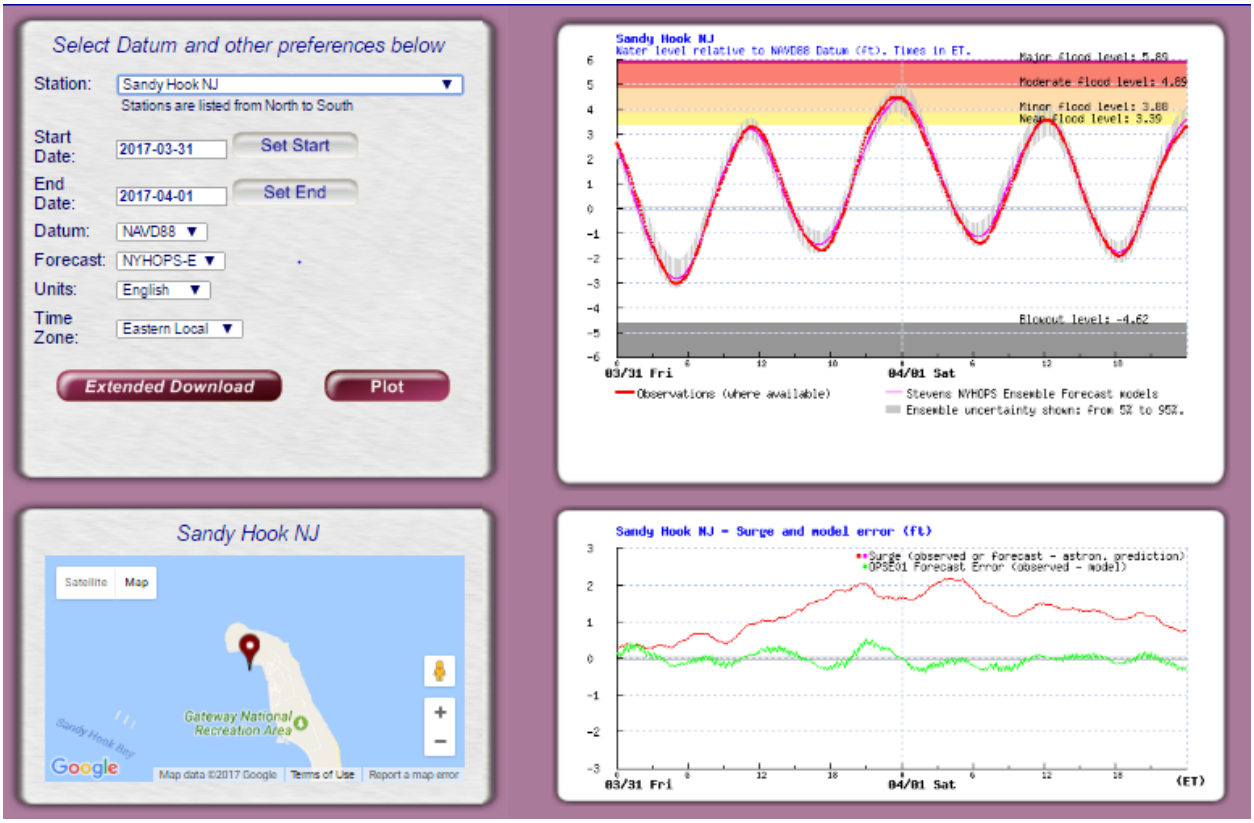


Kings Point NY

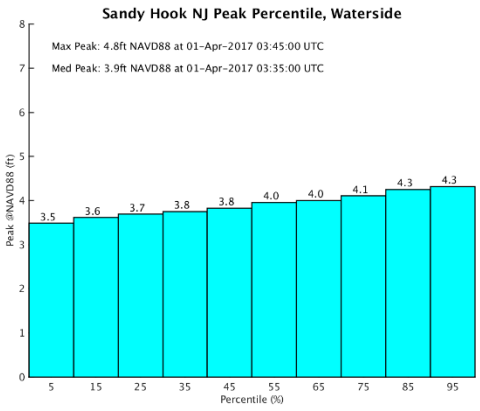


Percentiles of 96hr-prior forecast peaks.

Verified peak was 7.04ft NAVD88 on April 01 2017, 7:18 UTC.

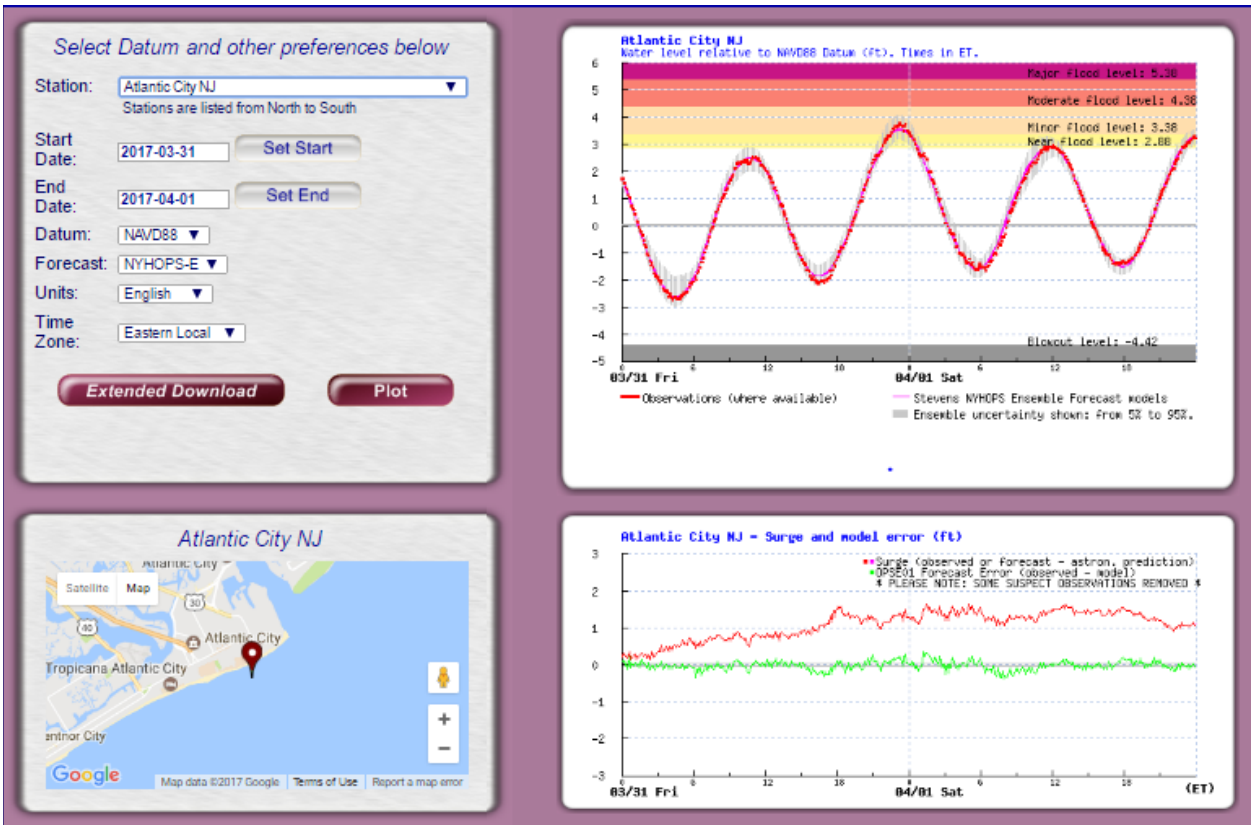


Sandy Hook NJ

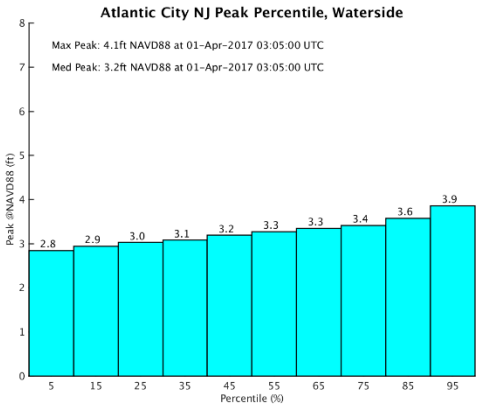


Percentiles of 96hr-prior forecast peaks.

Verified peak was 4.52ft NAVD88 on April 01 2017, 03:12 UTC.

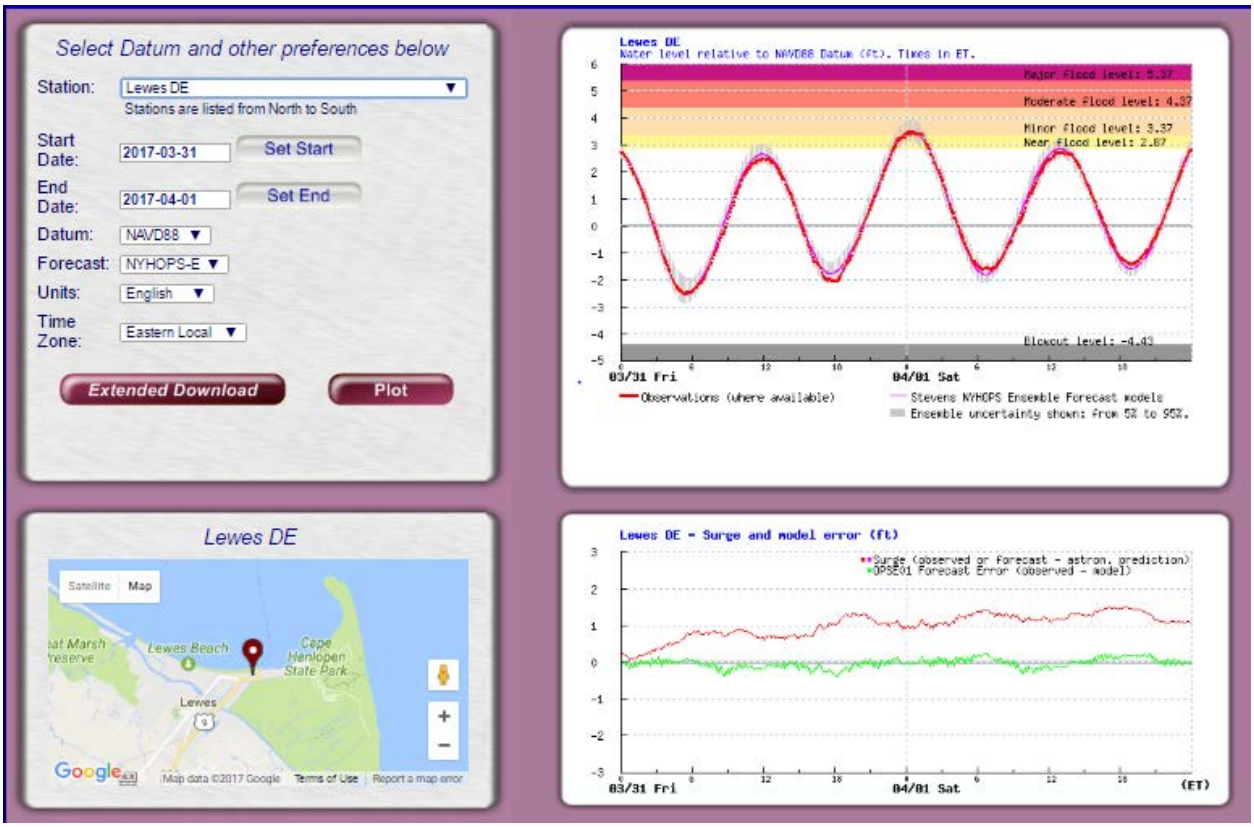


Atlantic City NJ

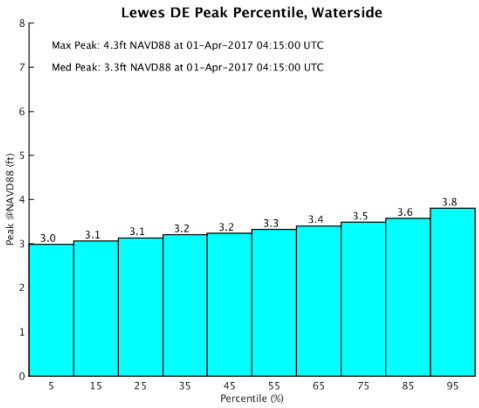


Percentiles of 96hr-prior forecast peaks.

Verified peak was 3.76ft NAVD88 on April 01 2017, 3:12 UTC.

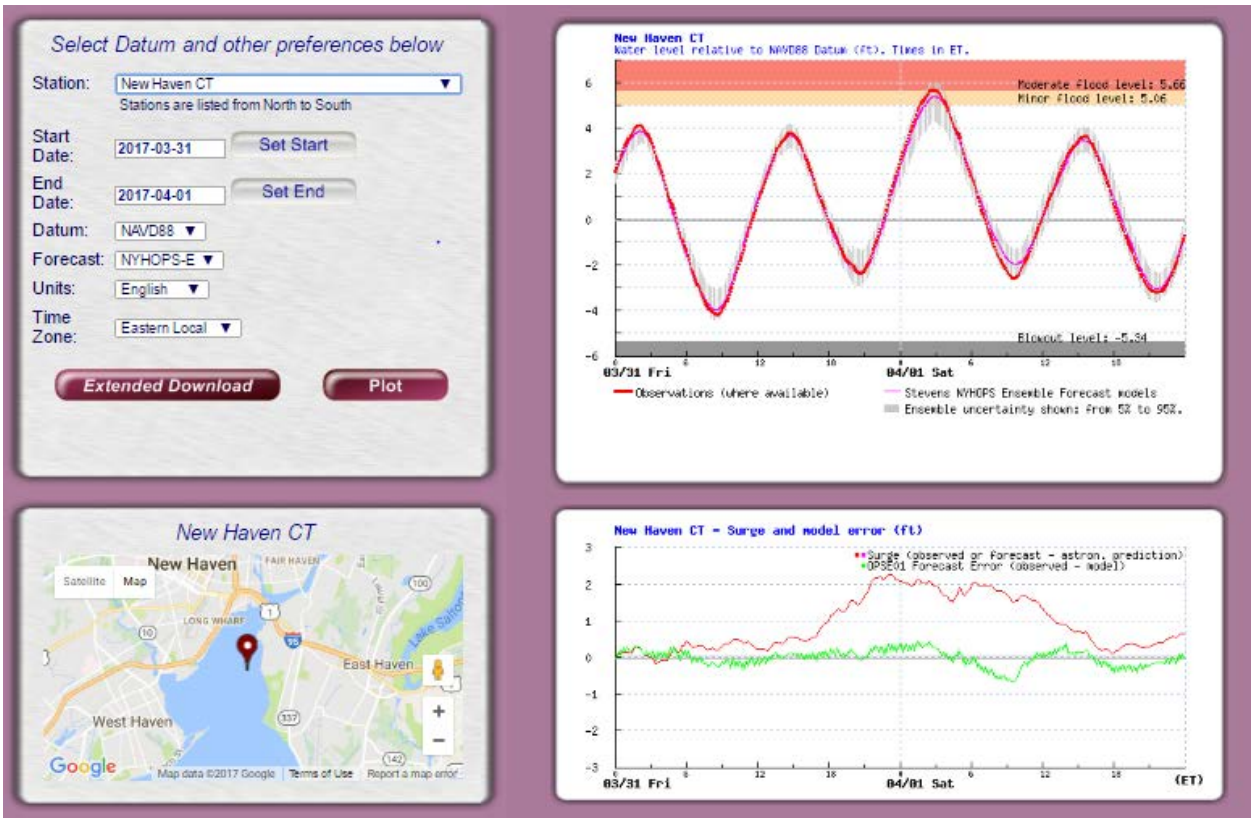


Lewes DE

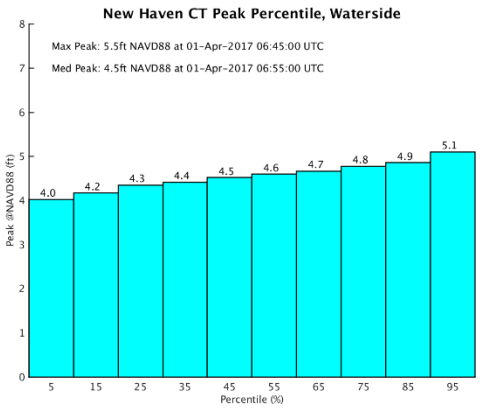


Percentiles of 96hr-prior forecast peaks.

Verified peak was 4.35ft NAVD88 on April 01 2017, 04:12 UTC.

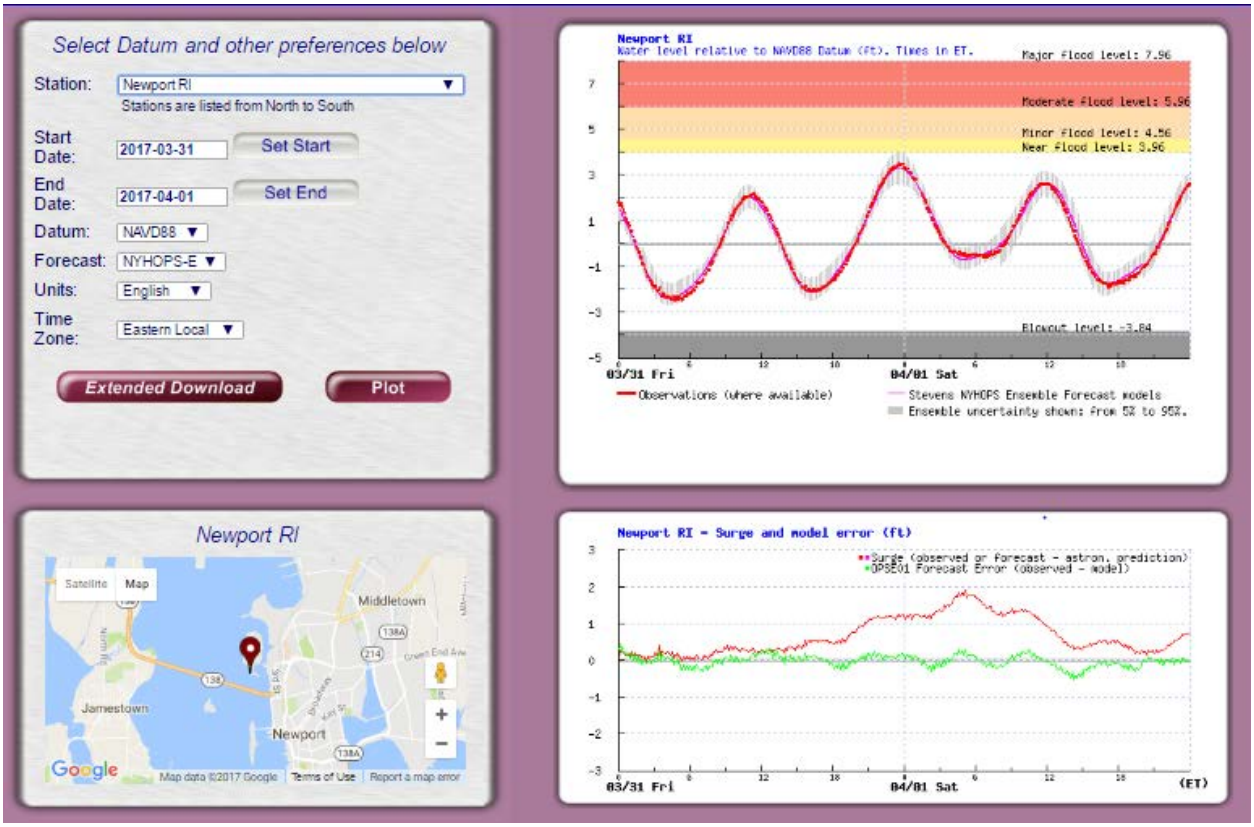


New Haven CT

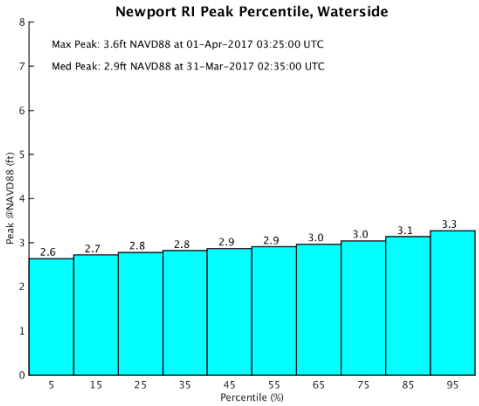


Percentiles of 96hr-prior forecast peaks.

Verified peak was 5.68ft NAVD88 on April 1 2017, 06:30 UTC.

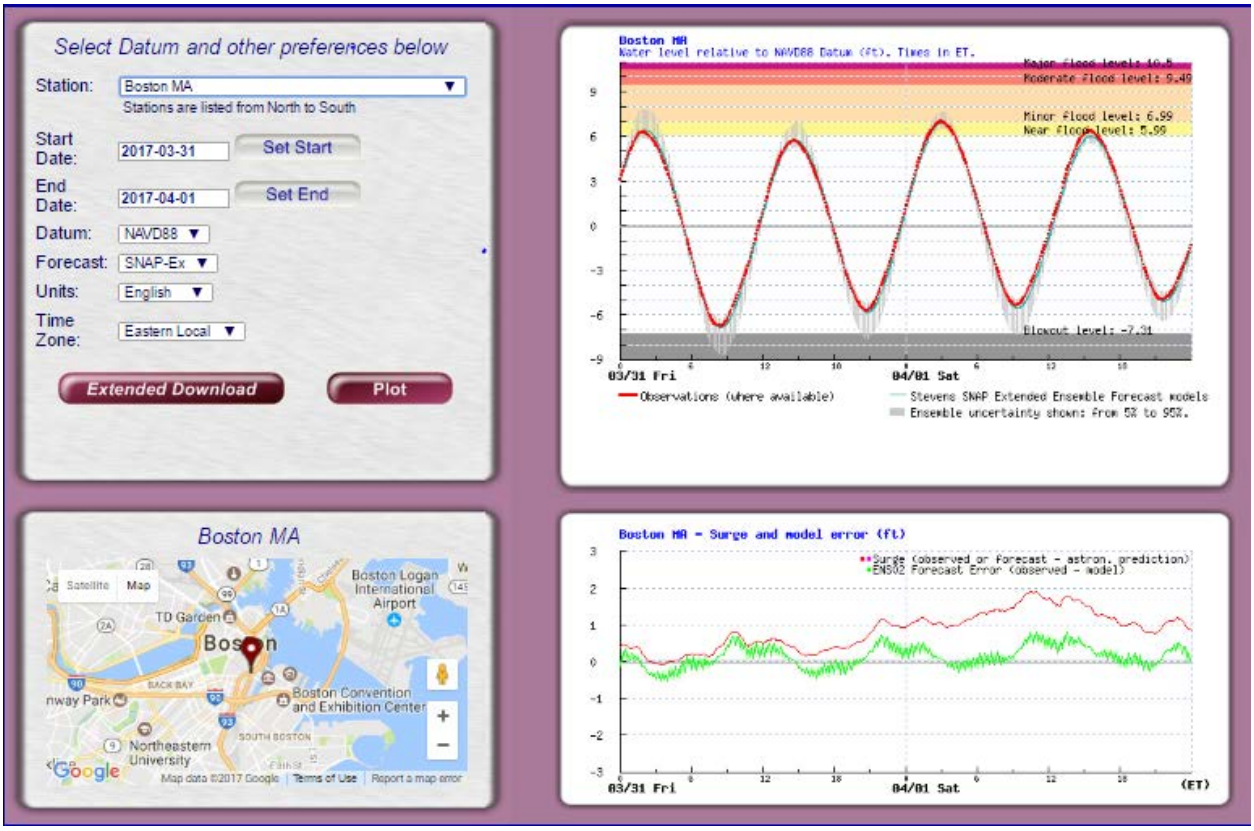


Newport RI



Percentiles of 96hr-prior forecast peaks.

Verified peak was 3.50ft NAVD88 on April 01 2017, 3:30 UTC.



Boston MA

Finally, Table 3 (next page) shows the observed or, in the absence of observations, simulated peaks for the storm.

Table 3. Preliminary Peak Water Levels for the Spring Tidal Event of April 1st 2017 at Stevens Flood Advisory System Stations

Station Name	Peak Water Level (ft NAVD88)	Peak time (EDT)	NWS Flood Level
Portland ME	6.45	4/1/2017 2:54	Near
Fort Point NH	6.28	4/1/2017 3:00	Near
Boston MA	7.06	4/1/2017 2:54	Minor
Providence RI	3.93	3/31/2017 23:42	Near
Newport RI	3.5	3/31/2017 23:30	
Woods Hole MA	2.42	4/1/2017 0:12	
Nantucket MA	2.84	4/1/2017 3:54	
New London CT	3.21	4/1/2017 1:24	Minor
New Haven CT	5.68	4/1/2017 2:30	Moderate
Bridgeport CT	4.77	4/1/2017 1:12	
Hudson River at Albany NY	7.03	4/1/2017 8:00	
Schodack Island NY	6.31	4/1/2017 7:30	
Norrie Point NY	4.08	4/1/2017 5:00	Minor
Hudson River at Piermont NY	4.04	4/1/2017 2:00	Near
Hackensack River at Hackensack NJ	4.75	4/1/2017 1:54	Minor
Barge Marina at Carlstadt NJ	5.73	4/1/2017 2:00	Minor
Passaic River near Harrison NJ	4.7	4/1/2017 0:36	
Passaic River at PVSC at Newark NJ	4.67	4/1/2017 0:30	Minor
Elizabeth Channel near Newark NJ	4.73	4/1/2017 0:12	
Newark Bay near Elizabeth NJ	4.5	4/1/2017 0:12	
Arthur Kill near Bloomfield NY	4.72	4/1/2017 0:25	
Bergen Point West Reach NY	4.59	4/1/2017	Minor
Kearny Point at Kearny NJ	4.63	4/1/2017 1:00	Minor
New York Harbor near Bayonne NJ	4.25	4/1/2017 0:30	
Hudson River near Hoboken NJ	4.22	4/1/2017 0:36	Near
Hudson River near Hudson Square NY	4.21	4/1/2017 0:42	

The Battery NY	4.45	3/31/2017 23:54	Near
New York Harbor near Red Hook NY	4.04	4/1/2017 0:36	
New York Harbor near Bay Ridge NY	4.31	3/31/2017 23:48	
Bowery Bay at East Elmhurst NY	6.63	4/1/2017 3:30	Minor
Kings Point NY	7.04	4/1/2017 3:18	Moderate
Bergen Basin at Jamaica Bay NY	4.27	3/31/2017 23:36	Near
Jamaica Bay at Inwood NY	4.56	3/31/2017 23:30	Minor
East Rockaway Inlet at Atlantic Beach NY	4.01	3/31/2017 23:42	Near
Reynolds Channel at Point Lookout NY	3.86	3/31/2017 23:30	Near
Great South Bay at Lindenhurst NY	2.06	4/1/2017 0:18	Near
Montauk NY	2.44	4/1/2017 0:45	
Raritan Bay at Keansburg NJ	4.85	3/31/2017 23:24	Minor
Sandy Hook NJ	4.52	3/31/2017 23:12	Minor
Shrewbury River at Sea Bright NJ	4.13	4/1/2017 0:54	Minor
Shark River at Belmar NJ	4.22	3/31/2017 23:36	Moderate
Watson Creek at Manasquan NJ	3.85	3/31/2017 23:36	Moderate
Barnegat Bay at Mantoloking NJ	1.2	4/1/2017 15:30	
East Thorofare at Ship Bottom NJ	2.53	4/1/2017 2:18	Minor
Little Egg Inlet near Tuckerton NJ	3.05	3/31/2017 23:54	Minor
Absecon Creek Rte 30 at Absecon NJ	3.58	4/1/2017 1:06	Minor
Inside Thorofare Rte 40 Atlantic City NJ	3.73	4/1/2017 0:36	Minor
Atlantic City NJ	3.76	3/31/2017 23:12	Minor
Ludlam Thorofare at Sea Isle City NJ	3.59	4/1/2017 0:24	Near
Great Channel at Stone Harbor NJ	3.52	4/1/2017 0:30	Near
Cape May NJ	4.01	4/1/2017 0:18	Minor
Maurice River at Bivalve NJ	4.19	4/1/2017 0:48	Minor
Lewes DE	3.48	4/1/2017 0:12	Minor
Ocean City Inlet MD	1.8	3/31/2017 23:48	
Cambridge MD	1.85	3/31/2017 19:18	
Annapolis MD	1.57	3/31/2017 20:42	Near

Baltimore MD	1.74	3/31/2017 9:24	Near
Lewisetta VA	1.75	4/1/2017 5:12	Near
Windmill Point VA	1.35	4/1/2017 2:06	
Barnegat Bay at Barnegat Light NJ	2.65	3/31/2017 23:18	Near
Chesapeake Bay Bridge Tunnel VA	2.29	3/31/2017 23:54	
Duck NC	2.54	4/1/2017 23:42	
Tivoli Bays South NY	4.94	4/1/2017 6:15	Minor
Hudson River below Poughkeepsie NY	4.4	4/1/2017 4:25	Minor
Hudson River at South Dock at West Point NY	4.08	4/1/2017 3:35	Minor
Hudson Rv South of Hastings-on-Hudson NY	4.28	4/1/2017 1:45	Near
FDU Bridge at Hackensack NJ	5.82	4/1/2017 2:35	Minor
Saw Mill Creek at Kearny NJ	4.86	4/1/2017 1:45	Minor
Hudson River near Jersey City NJ	4.39	4/1/2017 0:45	Minor
Hudson River at 39th St NY	4.31	4/1/2017 0:55	
Raritan Bay at South Amboy NJ	4.87	4/1/2017 0:05	Minor
Rockaway Inlet near Floyd Bennett Field NY	4.43	3/31/2017 23:55	
Hudson Bay at Freeport NY	3.74	3/31/2017 23:18	Minor
Peck Bay at Ocean City NJ	3.14	4/1/2017 0:05	Near
Cohancey River at Greenwich NJ	3.59	4/1/2017 2:42	Minor